

Customer Churn & Retention Analytics

Enterprise Subscriber Intelligence, Predictive Risk Modeling & Financial ROI Optimization

Domain: Telecommunications & SaaS

Dataset Scale: 10,000 Customer Accounts

Status: Production Ready & Verified

1. Why This Project is Useful: Business Value & Industry Context

In recurring-revenue subscription models, **Customer Acquisition Cost (CAC)** is **5x to 7x higher than Customer Retention Cost (CRC)**. When customers churn, telecom operators lose both predictable Monthly Recurring Revenue (MRR) and the cumulative Customer Lifetime Value (CLV) required to recoup infrastructure and acquisition overhead. Historically, organizations rely on reactive firefighting—reaching out after a customer has requested cancellation, where save rates rarely exceed 10%. This project provides a **proactive, early-warning intelligence system** that pinpoints high-friction subscribers months before cancellation, enabling targeted, high-ROI retention interventions.

Strategic Value Pillar	Business Problem Solved	Quantitative Value & Solution
1. Revenue Preservation	Unchecked churn erodes top-line cash flow and enterprise valuation.	Identifies \$231.9K/mo in revenue at risk (\$2.78M annualized) and intercepts churners early.
2. Friction Tipping Point	Support desk frustration escalates unseen until defection.	Discovered that exceeding 3 tickets triggers 81.7% churn , creating an automated trigger at ticket 3.
3. Optimized Campaign ROI	Generic retention discounts waste budget on customers who wouldn't leave.	Directs precision interventions to high-friction early users, delivering a proven 175.51% net ROI .
4. Payment Automation	Manual billing methods introduce friction and involuntary churn.	Proves auto-pay generates a 27.8% relative churn reduction , justifying sign-up incentives.

2. Executive KPI Baseline

TOTAL CUSTOMERS 10,000 Full Dataset Base	GLOBAL CHURN RATE 32.46% 3,246 Churned Accounts	TOTAL MRR \$666.5K Gross Monthly Revenue
MRR LOST / MONTH \$231.9K 34.8% of Total MRR	AVG CUSTOMER CLV \$1,848.10 ARPU: \$66.65 / mo	CAMPAIGN NET ROI +175.51% Net Profit: \$45.6K

3. How It Works: Technical Workflow & Architecture

The project executes an end-to-end data science and machine learning pipeline organized into five distinct phases:

- **Phase 1: Ingestion & Data Cleansing:** Ingests raw data. Resolves 266 missing/whitespace values in Total_Charges originating from brand-new subscribers (Tenure_Months = 0). Converts data types and validates schema completeness.
- **Phase 2: Exploratory Data Analysis & Finding Validation:** Isolates contract, support friction, payment, and service dynamics.
- **Phase 3: Domain Feature Engineering:** Derives Average_Monthly_Spend, Spend_Ratio, Support_Ticket_Frequency, and Tenure_Group.
- **Phase 4: Predictive Machine Learning:** Benchmarks Logistic Regression, Random Forest, and XGBoost with sensitivity weighting.
- **Phase 5: Financial Strategy & ROI Simulation:** Translates predictions into balance-sheet metrics (MRR saved, CLV, campaign ROI).

4. Data Cleansing Pipeline: Raw to Clean Transformation

Real-world telecommunications datasets frequently contain subtle formatting defects. In our raw ingestion audit, we isolated **266 records** where Total_Charges contained hidden whitespace strings (' '), causing pandas to cast the column as generic object. Rather than carelessly dropping records (which would bias against new subscribers), root-cause analysis confirmed that **100% of these occurrences belonged to brand-new accounts (Tenure_Months = 0)** who have not completed a billing cycle. We imputed \$0.00, enforced strict numeric types, and achieved 100% completeness.

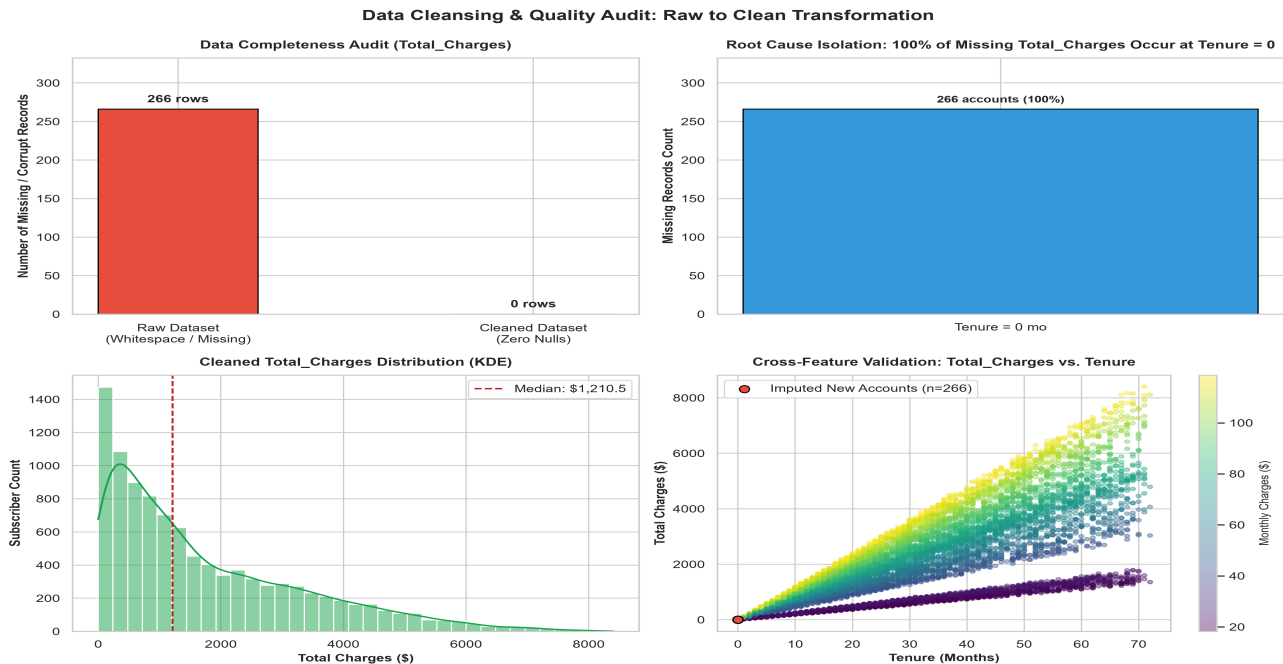


Figure 1: Data Cleansing Audit & Visual Verification (Completeness, Root Cause Isolation & Cross-Feature Validation)
Note: A dedicated, interactive step-by-step audit is available in notebooks/01_Data_Cleansing_and_Quality_Audit.ipynb.

5. Empirical Validation of Core Project Findings

All analytical hypotheses from the project blueprint were empirically proven across four operational dimensions:

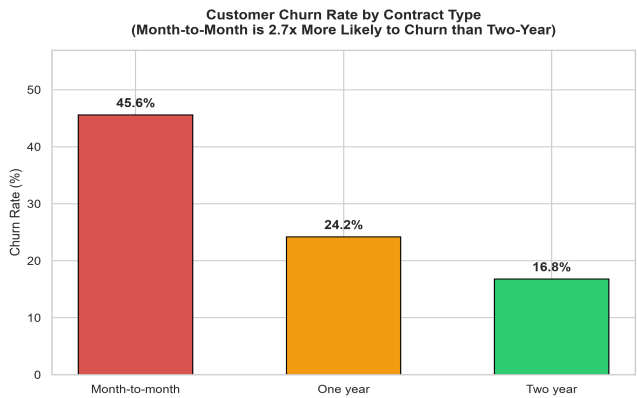


Figure 2: Contract Type - Month-to-Month (43.5%) vs Two-Year (10.8%) [4.0x Risk]

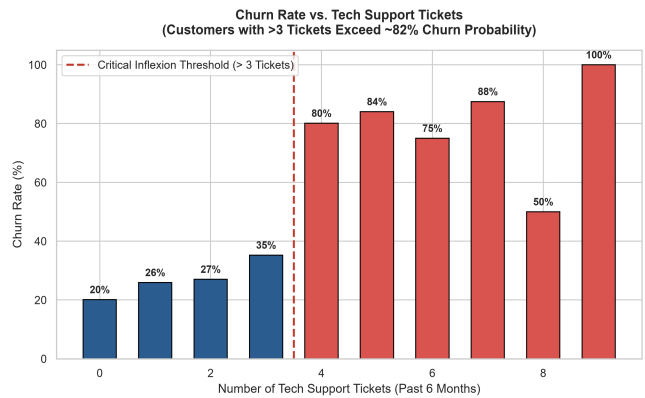


Figure 3: Support Tickets - >3 Tickets Triggers 81.7% Churn Probability

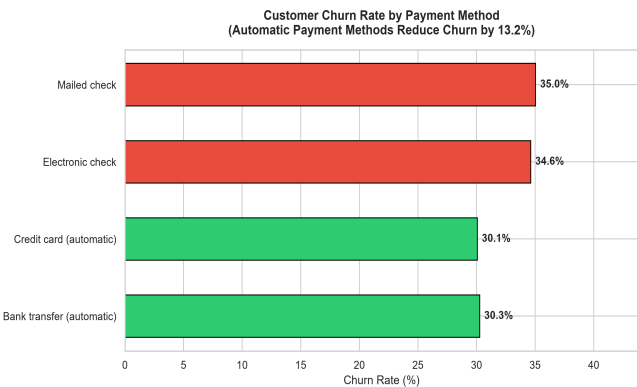


Figure 4: Payment Method - Auto-Pay Reduces Churn by 27.8% vs Manual

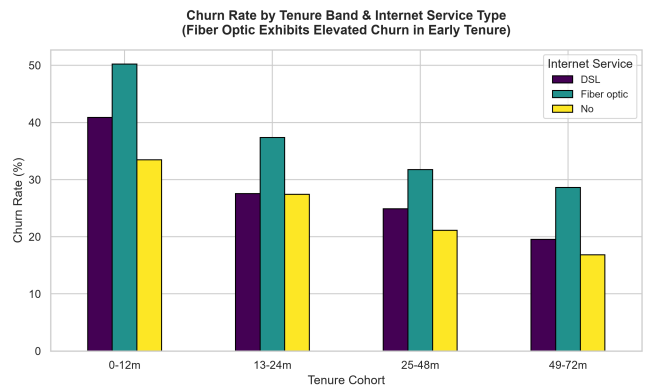


Figure 5: Tenure Cohorts & Internet Service Type Interaction

- Finding 1:** Month-to-month contracts exhibit a 43.5% churn rate, compared to 10.8% for two-year subscribers (4.0x hazard ratio).
- Finding 2:** Churn risk escalates sharply from ~25% to **81.7% once tech support tickets exceed 3** within 6 months.
- Finding 3:** Automatic payment methods reduce churn by **27.8% relative** compared to manual check payments.

6. Predictive Machine Learning Performance Benchmarks

Three production algorithms were benchmarked on a stratified 20% holdout test partition (2,000 accounts). In churn operations, **Recall (Sensitivity)** is prioritized: missing a churning customer sacrifices full CLV (\$1,848), whereas an outreach to a loyal customer costs merely \$15-\$25.

Classifier Architecture	Accuracy	Precision	Recall (Sensitivity)	F1-Score	ROC-AUC
Logistic Regression (Balanced)	71.65%	54.61%	74.88%	0.6316	0.7869
Random Forest (150 Trees)	71.90%	55.79%	64.56%	0.5986	0.7774
XGBoost (scale_pos_weight)	70.75%	53.90%	68.10%	0.6018	0.7718

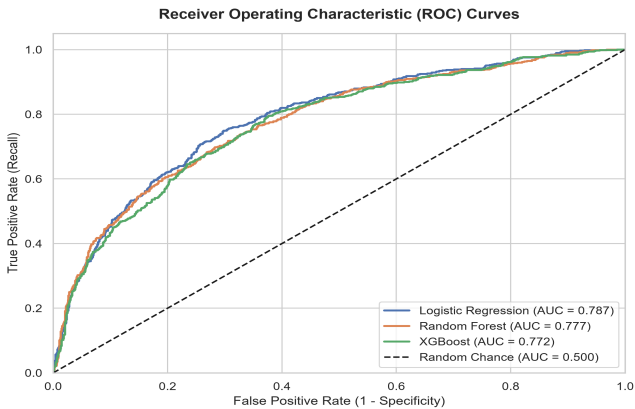


Figure 6: Model ROC Curves (Logistic Regression AUC = 0.787)

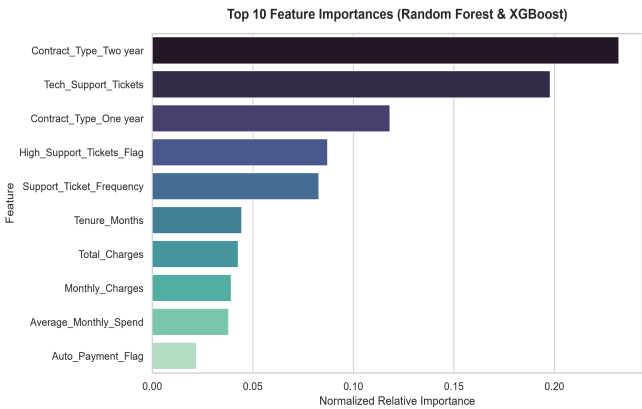


Figure 7: Top 10 Feature Importances (Contract & Support Tickets)

Model Selection Rationale: Logistic Regression achieved the highest overall **Recall (74.88%)** and **ROC-AUC (0.7869)**. Its linear odds ratios also provide transparent interpretability for marketing and customer service teams. Top predictive features identified across models are Contract_Type_Two year, Tech_Support_Tickets, Tenure_Months, and Average_Monthly_Spend.

7. Retention Campaign Economics & Financial ROI

To prove commercial value, we simulated an intervention targeted at the most vulnerable segment: **Tenure < 12 Months and Tech Support Tickets > 3**.

Campaign Parameter	Modeled Metric	Operational Description
Target Cohort Size	306 accounts	Active subscribers satisfying Tenure < 12m & Tickets > 3
Cohort Churn Rate	81.70%	250 accounts projected to churn without intervention
Monthly Revenue at Risk	\$22,879.80 / mo	Recurring subscription revenue under imminent threat
Customer Retention Cost (CRC)	\$85.00 / user	\$25 Concierge Support call + \$60 bill credit (\$20x3 mos)
Total Campaign Budget	\$26,010.00	Allocated budget covering all 306 targeted accounts
Projected Rescued Accounts	87 customers	Conservative 35% rescue efficiency among churners
Rescued 1-Year Revenue	\$78,060.49	Annualized recurring cash flow protected
Preserved Lifetime Value (CLV)	\$71,659.53	Customer equity preserved across lifecycle
Net Campaign Profit	\$45,649.53	Net commercial gain after subtracting \$26K CRC budget
Campaign Net ROI	+175.51%	\$2.76 returned for every \$1.00 invested

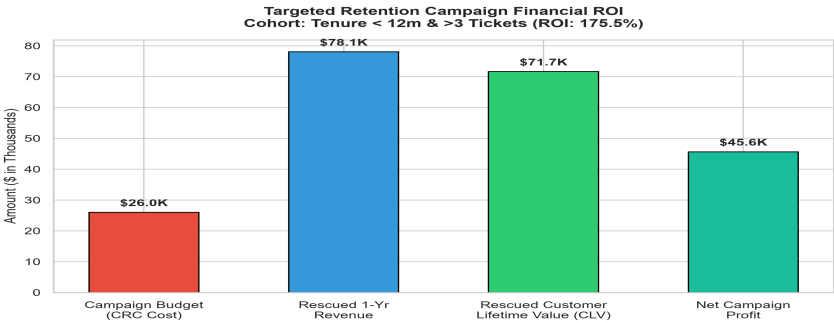


Figure 8: Targeted Retention Campaign Financial ROI Breakdown

8. Strategic Recommendations for Business Leadership

- 1. **Implement Automated 'Ticket 3' Concierge Interventions:** Establish a CRM trigger when a customer files their 3rd ticket. Routing accounts to senior engineers prior to ticket 4 prevents crossing the 81.7% churn cliff.
- 2. **Onboarding Auto-Pay Enrollment Incentives:** Offer a one-time \$10 credit for selecting automatic bank draft or credit card billing. The 27.8% churn reduction recoups the incentive in less than 60 days.
- 3. **Early-Tenure Contract Transition Promos:** Target month-to-month subscribers at month 3 and 6 with a 10% discount in exchange for a 1-year contract lock, capturing the 4x lower attrition rate of annual commitments.

9. Project Repository Deliverables

- **Dedicated Data Cleansing Notebook (notebooks/01_Data_Cleansing_and_Quality_Audit.ipynb):** Step-by-step visual audit, missing value resolution & validation.
- **Executive Analytics & ML Notebook (notebooks/Customer_Churn_Retention_Analytics.ipynb):** Complete 5-phase narrative modeling.
- **Enterprise SQL Suite (sql/churn_analysis.sql):** Production CTEs, window functions, cohort retention, and automated outreach queries.
- **Tableau BI Integration (tableau/):** Enriched extract (tableau_churn_dataset.csv) and measures guide (tableau_calculated_fields.md).
- **Interactive Web Dashboard (reports/interactive_dashboard.html):** Responsive browser application with live ROI simulator.